Melika Baghi

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Summary

Ph.D. Candidate in Industrial & Systems Engineering at Georgia Tech with an M.S. in Statistics. Work covers experiment design, ETL and feature pipelines, predictive modeling and statistical validation, through to production-oriented pipelines. Seeking data scientist and applied machine learning roles.

Technical Skills

Programming & data: Python, SQL, PySpark, R, scikit-learn, pandas, NumPy, SciPy, Airflow, MLflow
Machine learning: scikit-learn, XGBoost, PyTorch, ensemble methods, deep learning, representation and multimodal learning
Statistics: experiment design, hypothesis testing, nonparametric methods, calibration, uncertainty quantification, causal and time-series modeling
Delivery: ETL pipelines, model tracking, dashboards (Power BI), Git, Docker, Linux, GPU/HPC, cloud computing (AWS, Azure)

Projects

- **Hospital capacity and patient flow** — hybrid agent-based and discrete-event simulation of a hospital serving two patient populations; a 256-run fractional factorial design over 16 operational factors found bed capacity the most consistent driver of waiting, and without patient behavior the model's admission wait rose from **14.8 to 25.8 days** ($p < 0.001$). *AnyLogic, Python*
- **Predictive maintenance at scale** — end-to-end pipeline for sensor-based fault detection. ARIMA and LSTM forecasting with ensemble anomaly classifiers over multivariate telemetry gave over **15 hours** of advance warning; deployed on Spark for low-latency inference. *Python, PySpark*
- **End-to-end ML pipelines** — CRISP-DM workflows over simulation, sensor and transactional data. ETL in PySpark and SQL, model tracking automated with Airflow and MLflow, results delivered as dashboards in Python, R and Power BI.
- **Clinical risk from EHR** — adverse drug event prediction from longitudinal records, compared across regularized regression, tree ensembles and neural models, with calibration and fairness checks across demographic groups. *Python*
- **Credit risk modeling** — ensemble classifiers for consumer loan default, evaluated on ranking performance and probability calibration rather than accuracy alone. *Python*
- **Cross-national statistical inference** — nonparametric study of internet access against female employment and management representation across 92 countries; Mann–Whitney tests with Bonferroni correction and Anderson–Darling normality diagnostics. *Python*
- **Anomaly detection in sensor streams** — ARMA-based statistical monitoring with sliding-window control charts to flag faults early, trading off detection delay against false-alarm rate. *Python*
- **Breast cancer prediction** — diagnostic classification pipelines compared on held-out generalization rather than training accuracy. *Python, scikit-learn*
- **Multimodal phenotype modeling** — contrastive representation learning over single-cell transcriptomics, longitudinal EHR and imaging-derived phenotypes, with transformer language models extracting structured phenotypes from unstructured clinical notes. *Python, PyTorch*
- **Constrained gradient boosting (cXGBoost)** — enforces a manifold validity bound inside every leaf during fitting, cutting reconstruction error 5× against interpolation baselines; released as an open-source Python package. arXiv:2605.04130
- **Tensor-structured surrogates** — keeps the multiway structure of simulation output instead of flattening it: 13×–482× smaller interpolation size and 3.3×–8.5× faster than flattened baselines across five PDE benchmarks, within 0.22 percentage points of their accuracy. Submitted to J. Comput. Phys.
- **Prediction under missing modalities (RouteCert)** — certifies a prediction when the modalities already collected determine the decision and otherwise recommends which to acquire next, with distribution-free coverage guarantees; evaluated on ECG, speech and wearable-sensing datasets. arXiv:2608.15520

Education

Ph.D., Industrial & Systems Engineering, Georgia Institute of Technology *Aug 2023 – Dec 2027 (expected)*
M.S., Statistics, Georgia Institute of Technology *Aug 2023 – May 2026*
M.S., Industrial and Systems Engineering — Healthcare Systems, Amirkabir University of Technology; GPA 4.0/4.0 *2021 – 2023*
B.S., Industrial Engineering, Amirkabir University of Technology; GPA 3.91/4.0 *2018 – 2021*

Publications: five first-author arXiv preprints (2610.34559, 2608.15520, 2609.01761, 2605.04130, 2605.19139); full list.